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Semi-supervised generative adversarial network for plant leaf disease detection

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ABSTRACT

Fully supervised plant leaf disease segmentation methods based on convolutional neural networks (CNN) require large quantities of labeled images for training, which are time-consuming and labor-intensive to obtain in practical scenarios. Moreover, most diseases of plant leaf exhibit indistinct edge information and background noise interference, which hinders precise disease localization. To overcome these issues, a semi-supervised generative adversarial network (SSGAN) is proposed for plant leaf disease inspection. Firstly, to learn comprehensive semantic features from limited data, a semi-supervised method based on generative adversarial networks is developed to enhance the interaction of features among labeled and unlabeled disease images. Secondly, a boundary feature attention module (BFAM) is proposed to enhance edge detail representation, which makes the model pay more attention to the boundary feature of plant leaf diseases. Finally, a background noise suppression module (BNSM) is proposed to bolster the differentiation between the normal and disease areas so as to alleviate the adverse impact of background noise. The effectiveness of SSGAN is verified on two plant leaf disease datasets. The mean intersection over union (mIoU) on the apple and tomato leaf datasets improves by 2.07 % and 2.28 % respectively, compared with the suboptimal method. The testing results of experiments show that SSGAN achieves great performance on plant leaf disease segmentation under limited labeled data.

1. Introduction

Plant leaf diseases can significantly impair crop yields, which degrades the quality of crop production. Thus, timely inspection of plant leaf diseases is important to maintain plant health and reduce crop loss (Gupta and Tripathi, 2024) (Sharma et al., 2022). Conventional approaches to detecting plant diseases primarily depend on visual inspection and manual diagnosis, which is not only laborious but also constrained by the expertise and proficiency of the individuals involved (Panigrahi et al., 2020). As automated technology develops, some advanced methods have been developed for plant disease detection. For instance, Huang et al. (2025) proposed a triangular stable structure chemical product to improve the quality of plants. Luo et al. (2022a) developed a deep belief model to construct a monitoring system, which utilizes the Internet of Things to acquire agricultural information. Mukherjee et al. (2025) extracted integrated features from imagery, and then utilized classification algorithms like support vector machines for the training of their classification models. Yu et al. (2021) utilized random forest and linear regression to evaluate the health status of the corn canopy. Nevertheless, these methodologies typically necessitate a large quantity of labeled data to train a model, which is impractical in real-world scenarios.

Recently, deep learning-based methods have achieved remarkable performance in plant leaf disease detection. For example, Ye et al. (2024) proposed a model for the detection of plant disease, which focuses on optimizing model parameters to promote precision. Yu et al. (2023) developed an ultrasound-based method that incorporates a denoising filter for monitoring plant health conditions. Shi et al. (2017) proposed a disease inspection technique grounded in dimensionality reduction, which has high performance in detecting apple leaf diseases.

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Karthik et al. (2020) leveraged transfer learning to identify and categorize a range of tomato leaf diseases, employing an attention mechanism to bolster the robustness of the model across diverse plant disease datasets. Sharma et al. (2025) developed a transformer (Zhao et al., 2025) model for better detection of soybean leaf disease. Tu et al. (2024) proposed a graph frequency method to perform plant leaf disease detection in hyperspectral images, which makes full use of spectral information to assist in the location of anomalies. Nonetheless, annotating a substantial volume of plant leaf disease data remains a laborious and time-intensive task.

In recent years, there has been a shift towards semi-supervised approaches that concentrate on training with a limited set of annotated defective data alongside a vast quantity of unannotated data. Sharma et al. (Sharma and Sharma, 2024) proposed a semi-supervised model for diagnosing plant diseases, leveraging ensemble learning to attain precise detection even with a scarcity of labeled data. Li and Chao (2021) developed a weakly-supervised approach for identifying disease, employing confidence intervals to dynamically select unlabeled samples for generating pseudo-labels. Zhong et al. (2023) developed a groundbreaking technique for image segmentation, which utilizes labeled and unlabeled images to construct a framework of confidence among various disease characteristics.

Although these methods have demonstrated good performance in plant disease detection, several limitations persist: (1) These deep-learning-based methods for plant leaf disease detection predominantly rely on supervised learning, which necessitates a significant quantity of labels to cultivate a model that delivers high-performance outcomes. (2) The edge of plant diseases is mostly fuzzy, which presents difficulties for accurate disease localization. (3) Moreover, the background of disease images contains lots of noise and interference, which restricts the detection accuracy on plant leaf diseases.

In this study, a novel semi-supervised generative adversarial network (SSGAN) is proposed for plant leaf disease segmentation to overcome the problems of label scarcity and background noise. To summarize, the contributions of this article are presented as follows: (1) A novel semi-supervised model based on generative adversarial network (GAN) is proposed to overcome the problem of label scarcity, which enables the network to acquire comprehensive semantic features from unlabeled diseases images; (2) To enhance discriminative feature of leaf disease boundary, a boundary feature attention module (BFAM) composed of global and local attention is proposed to enable the network to generate boundary information representation associated with plant leaf diseases; (3) To alleviate the influence from background noise, background noise suppression module (BNSM) is proposed to bolster the differentiation between diseased area and background. The testing results on two leaf disease datasets demonstrate the effectiveness of SSGAN for leaf disease detection.

The remaining content of this paper is as follows. Section 2 presents related works for plant disease detection, semi-supervised learning, and Generative Adversarial Networks. The detailed presentation about SSGAN is presented in Section 3. Section 4 verifies the effectiveness of SSGAN for plant leaf disease detection. Finally, Section 5 concludes this study.

2. Related work

2.1. Plant leaf disease inspection

Automated plant leaf disease detection methods serve as potent instruments for the early and precise inspection of plant diseases, which can be divided into machine learning-based and deep learning-based methods. As for machine learning-based methods, Sia et al. (2025) utilized random forests and decision trees to obtain geometry features of plant leaf diseases, which can predict transcriptional information from the early growth stage. Das et al. (2022) developed a healthy monitoring method based on machine learning, which uses scale-invariant feature transform to analyze the genomic sequence of different plant leaf diseases. Sharma et al. (2021) proposed a color texture method for rice leaf disease detection, which utilized a support vector machine as a classifier to accurately detect leaf diseases with different colors. To alleviate the influence of various sizes and rotations, Thiagarajan et al. (2024) utilized a histogram of oriented gradients as a feature extractor to obtain local details of banana disease leaves with complex shapes. As for deep learning-based methods, Tulshan and Raul (2019) developed an innovative network for the autonomous recognition of plant diseases, which extracts critical features through the computation of the feature matrix from leaf images. Zhao et al. (2023a) developed a feature-aligned network for plant disease detection, which enhances performance by aligning labeled and unlabeled data. Chouhan et al. (2021) developed a novel model for recognizing plant leaf disease, featuring a dual attention mechanism and the capability for multiscale feature extraction, which mitigates the negative impact of complex backgrounds on recognition outcomes. Deng et al. (2022) developed a model to detect diseases by using federated learning, which captures multi-scale features through diverse convolutional layers. Shah et al. (2024) developed a groundbreaking identification model premised on an enhanced EfficientNet architecture for fruit disease detection, which incorporates a self-attention with feature enhancement to obtain more accurate results.

2.2. Semi-supervised learning

Deep learning techniques are typically categorized into supervised, semi-supervised, and unsupervised learning. Due to the significant effort and time required to label data, it is common that only a portion of the data can be labeled in practical scenarios. Consequently, semi-supervised models have garnered significant interest in recent years. For instance, Wang et al. (2023) developed a novel method to address the limited availability of training data by utilizing data augmented methods for defect detection. Lin et al. (2023) employed a consistency regularization, where the teacher network is trained to generate effective pseudo labels from unlabeled data, thereby aiding the student model in remaining useful life (RUL) prediction. Yin et al. (2021) proposed a novel confidence score derived from the distribution of loss, which selects reliable pseudo labels to enhance few-shot classification. Li et al. (2024) developed a tumor detection technique grounded on unified imbalanced semi-supervised consistency learning, addressing both class imbalance and semi-supervised scenarios simultaneously. To increase the diversity of limited samples, Zhao et al. (Zhao and Yu, 2025) proposed a data-augmented method by cropping the object in one image and pasting it in another, which facilitates the feature fusion between labeled and unlabeled images. Samadi et al. (2025) developed a method for novel class classification, which used a dynamic weight strategy to make full use of features from semi-supervised learning.

2.3. Generative adversarial network (GAN)

As a form of unsupervised methods initially developed for image generation tasks, GAN has shown impressive performance across various fields. For example, Sharma et al. (2023) developed a data augmentation method based on GAN to generate synthetic images of weeds, which utilizes federated learning to reduce the computation load of discriminators. Inspired by the architecture of the transformer (Liu et al., 2022), Zhao et al. (2023b) refined the conditional deep convolutional GAN by incorporating bypass connections and subsequently augmented the defective dataset with specified category labels. Nazki et al. (2020) addressed issues stemming from data scarcity in diagnosing tomato tree diseases and leveraged GAN to mitigate class imbalance. Shi et al. (2023) employed a distance adversarial network for data enhancement in the detection of industrial process anomalies under imbalanced data conditions, which achieves superior detection accuracy compared to non-enhanced datasets. Zhang et al. (2021) substituted the generator of a traditional GAN with a semantic segmentation network and the

discriminator with a fully convolutional discriminant network, bypassing inherent GAN challenges and demonstrating effective performance in strip defect detection. Xu et al. (2022) developed a GAN-based method to remove highlights from the grey image, which uses a generator to provide highlight information and utilizes an attention block to detect the highlight region. Sharma et al. (2024) developed a novel GAN model to perform leaf disease detection, which employs an encoder-decoder network to avoid the vanishing gradient problem.

3. Methodology

3.1. Overview

In this part, a novel semi-supervised generative adversarial segmentation method, i.e., SSGAN, is proposed for plant leaf disease detection. The structure of SSGAN is shown in Fig. 1. SSGAN consists of two components: semi-supervised learning and generative adversarial learning. The processes can be described as follows: First, for semi-supervised learning, both labeled and unlabeled plant leaf disease images are fed into the encoder and decoder to generate predictions and pseudo labels, respectively. The encoder network is composed of ResNet34, BNSM and BFAM. The Atrous Spatial Pyramid Pooling (ASPP) module (Lian et al., 2021) is tasked with this role within the decoder. The rationale for using ASPP as a decoding module is as follows: Firstly, ASPP adopts convolutional layers with different dilation rates to obtain multi-scale features. Secondly, Atrous convolutions in ASPP expand the receptive field without reducing resolution, which enables the decoder to accurately locate and segment objects. Thirdly, ASPP acquires and integrates global information through global average pooling, which enables the decoder to understand the context feature. The encoder and decoder network can be seen as a generative network. Next, in terms of generative adversarial learning, the input of the discriminator is generated by concatenating the feature map and the image along the channel direction. Then, a confidence map is obtained by calculating the maximum probability of each pixel in the pseudo label, which generates a confidence score by average pooling. The confidence score will be compared with the threshold λ. If the confidence score is larger than λ, the generated image is judged as True. Otherwise, it will be judged as False. Finally, feature matching loss is conducted to update the parameters of SSGAN to make the generative network obtain accurate segmentation results.

3.2. Semi-supervised learning based on GAN

In plant leaf disease detection, it is time-consuming and labor-intensive to label large amounts of disease images. Additionally, a large number of semi-supervised techniques employ different learning frameworks, which causes the disregard of significant knowledge gained from the sparse labeled data. To address this issue, a semi-supervised approach based on GAN (Sharma et al., 2023) is developed to harness the strengths of both labeled and unlabeled data of diseases, which is shown in Fig. 1. For two images X_L and X_U that are labeled and unlabeled images, respectively, the prediction Y_L and pseudo label Y_U are generated as follows:

$$Y_L = SS(X_L); Y_U = SS(X_U) \tag{1}$$

where SS(.) denotes the encoder and decoder function. Next, cross-entropy loss is employed to calculate the loss between the result and label, which is formulated as follows:

$$L_{ce} = \sum (Y_L - G)^2 \tag{2}$$

where G represents the ground truth of labeled disease images, and L_{ce} denotes the cross-entropy loss. Then, feature matching loss (Cao et al.,

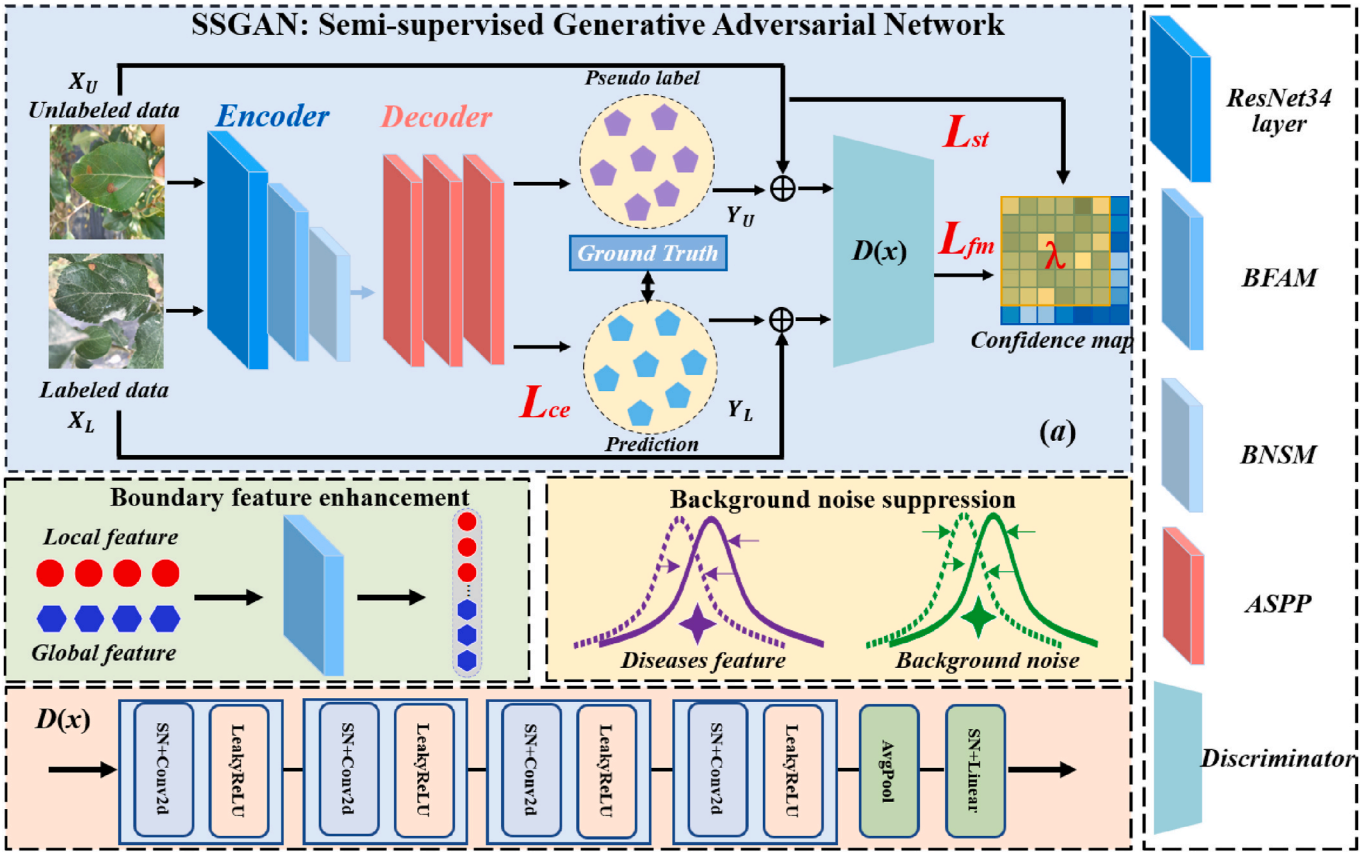


Fig. 1. Network structure of SSGAN.

2024) is employed to minimize the average difference between the result Y_U and the label G. The feature matching loss is formulated as follows:

$$L_{fm} = \| E_{(X_L,G)\sim D^l}[D_k(G \oplus X_L)] - E_{(X_U,Y_U)\sim D^u}[D_k(SS(X_U) \oplus X_U)] \| \tag{3}$$

where L_{fm} denotes feature matching loss, D denotes the discriminator, and k denotes the k-th layer in the discriminator. The feature matching loss is computed as follows: A specific layer of the discriminator D_k extracts features from two inputs: one is the combination of pseudo-labels and unlabeled images, which is from the output of the generator, and the other is the combination of ground-truth labels and labeled images. Then, the L1 norm of the expectations of these two types of features is calculated to minimize the distribution difference between the generator-related features and the real data features for the discriminator. During the training of GAN, it is crucial to maintain a balanced interplay between the generator and the discriminator. If the discriminator is too strong at the beginning of training, it will not provide any useful learning information for the generator. To promote this dynamic balance, self-training loss is considered to be added to the network. Its main idea is to select the optimal generator output for unlabeled data, which can deceive the output of the discriminator and reuse these outputs for pseudo-label self-training, which is defined as follows (Pu and Li, 2023):

$$L_{st} = \begin{cases} -\sum_{h,w,c} Y_U \log SS(X_U), if D(SS(X_U)) \geqslant \lambda \\ 0, otherwise \end{cases} \tag{4}$$

where λ represents the threshold.

3.3. Background noise suppression module

The noise interference from the background of the leaf disease images is a main reason that restricts the accuracy of the detection models. A background noise suppression module (BNSM) based on (Ge et al., 2025) is proposed to perform noise suppression to further improve the performance. Fig. 2 presents the architecture of the BNSM, which can be calculated as follows (Ge et al., 2025):

$$F_{out1} = F_{in1} + (F_{in1}\text{-A}) \times K(|F_{in1}\text{-A}|),\ F_{in1} \in \mathbb{R}^{C\times H\times W}, \text{A} \in \mathbb{R}^{C\times 1\times 1}, F_{out1} \in \mathbb{R}^{C\times H\times W} \tag{5}$$

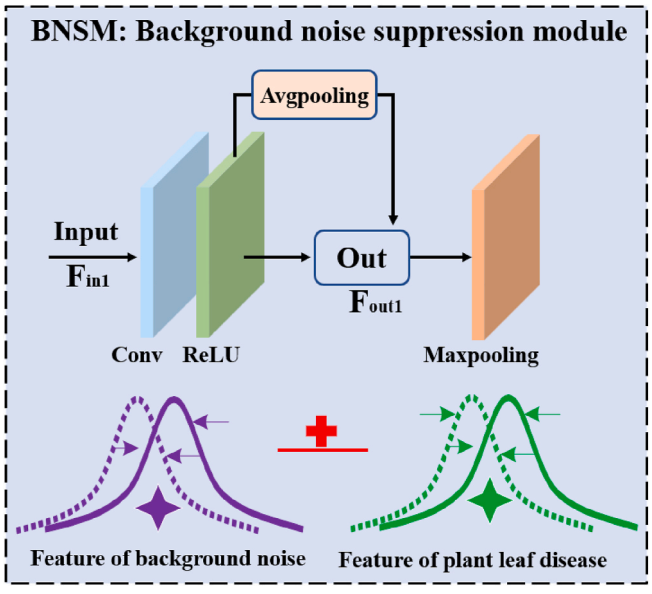


Fig. 2. Structure of background noise suppression module.

$$K(x) = \frac{1}{\pi} arctan(|x|) + \frac{1}{3}, x \in \mathbb{R}^{C\times H\times W} \tag{6}$$

$$\text{A} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \boldsymbol{I}_{ij}, \text{A} \in \mathbb{R}^{C\times 1\times 1},\ \boldsymbol{I}_{ij} \in \mathbb{R}^{C\times 1\times 1} \tag{7}$$

where F_{in1} and A denotes the input and average pooling result, respectively, F_{out1} denotes the outcome of BNSM, K (·) denotes the kernel algorithm, and $\boldsymbol{I}_{ij}$ denotes the element with the i-th row and the j-th column in the feature map. C, H and W represent the number of channels, height, and width of the feature map, respectively. The arctangent function in Eq. (6) is employed for its smooth transition and robust performance, which effectively suppresses background noise. Moreover, its non-linearity and normalization contribute to a balance between noise suppression and feature retention. To maintain the output value within an acceptable range, it is derived through the following process:

$$F_{out1} = Min(Max(F_{out1}, 0), 255), F_{out1} \in \mathbb{R}^{C\times H\times W} \tag{8}$$

As displayed in Fig. 2, the feature map generated by BNSM serves as the input immediately preceding the max pooling layer. In contrast to the unaffected regions of a plant leaf, disease-affected areas are typically smaller and exhibit distinct distribution patterns, which often causes the mean values to skew closer to those of the healthy regions. BNSM is proposed to mitigate this discrepancy by reducing the variance between the normal areas and the overall average. Additionally, integrating BNSM at the shallower layer aids in the extraction of disease-specific contour and texture information by the early convolutional layers.

3.4. Boundary feature attention mechanism

Since the boundary of diseases exhibits obscure feature information, both false-positive and false-negative predictions would naturally occur in the process of detection. It is necessary to obtain comprehensive features locally and globally. To solve this problem, boundary feature attention module (BFAM) composed of global and local attention is proposed, which is shown in Fig. 3. In particular, the input feature map F_{in2} is split into two feature maps F_1 and F_2, and then F_1 and F_2 are fed into global attention (GA) and local attention (LA), respectively. The results from the two modules are ultimately combined through concatenation, followed by a 1×1 convolution. The calculation process can be formulated as follows:

$$\begin{cases} F_1 = \begin{cases} F_{in2}(c,h,w), & c \leq C/2 \\ 0, & \text{otherwise} \end{cases} \\ F_2 = \begin{cases} 0, & c \leq C/2 \\ F_{in2}(c,h,w), & \text{otherwise} \end{cases} \end{cases}, F_1, F_2 \in \mathbb{R}^{\frac{C}{2}\times H\times W}; F_{in2} \in \mathbb{R}^{C\times H\times W} \tag{9}$$

$$F_{out2} = Conv_{1\times 1}(\text{Concat}(G_a(F_1), L_a(F_2))), F_{out2} \in \mathbb{R}^{C\times H\times W} \tag{10}$$

where c, h and w are the number of channels, height, and width of F_1 and F_2, respectively, C is the total number of channels in the input feature map F_{in2}, Concat represents the splicing operation of channel dimension, F_{out2} is the output feature map of BFAM, G_a represents global attention, L_a denotes local attention, and $Conv_{1\times 1}$ represents the 1×1 convolution. The G_a and L_a are as follows:

(1) Global attention: The Global Attention (GA) module accentuates the long-range dependencies among pixels across the spatial expanse and serves as a complementary enhancement to local attention, which can be formulated based on (Sharma et al., 2025) as follows:

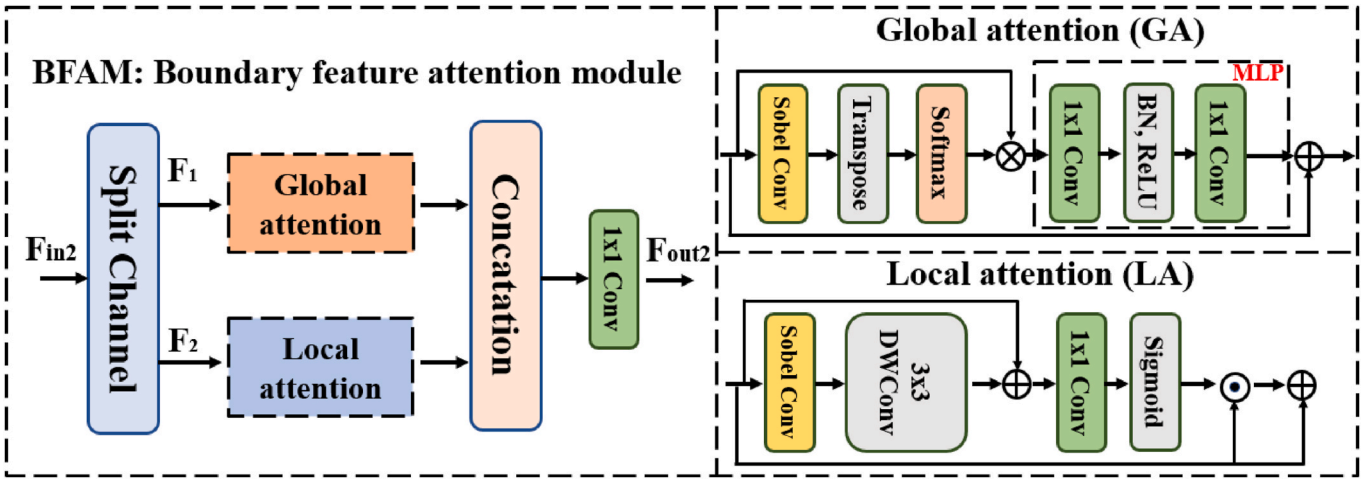


Fig. 3. Structure of boundary feature attention module.

$$Att_G(F_1)_{c,j,i}=\frac{\exp\left(\mathbf{S}^T_{c,j,i}\right)}{\sum_{c=1}^{\frac{C}{2}}\exp\left(\mathbf{S}^T_{c,j,i}\right)}, S\in\mathbb{R}^{\frac{C}{2}\times H\times W}, c=1,\ldots,\frac{C}{2}; j=1,\ldots,H; i=1,\ldots,W; \tag{11}$$

$$G_a(F_1)=MLP(Att_G(F_1)\otimes F_1)+F_1, F_1\in\mathbb{R}^{\frac{C}{2}\times H\times W} \tag{12}$$

where S is the output of Sobel Conv, $Att_G(\cdot)$ is the attention operation, $\otimes$ is matrix multiplication, and MLP(·) represents Multi-Layer Perceptron, T is the Transpose function. It should be noted that the MLP is composed of 1x1 Conv, Batch Normalization, ReLU, and 1x1 Conv. C, H and W represent the number of channels, height, and width of the feature map, respectively.

(2) Local attention: The Local Attention (LA) module efficiently captures the local features of the area from the perspective of spatial direction, which is formulated based on (Sharma et al., 2025) as follows:

$$Att_L(F_2)_{c,j,i}=\frac{1}{1+\exp\left(-\boldsymbol{S}_{c,j,i}\right)}, S\in\mathbb{R}^{\frac{C}{2}\times H\times W}, c=1,\ldots,\frac{C}{2}; j=1,\ldots,H; i=1,\ldots,W; \tag{13}$$

$$L_a(F_2)=Att_L(F_2)\odot F_2+F_2, F_2\in\mathbb{R}^{\frac{C}{2}\times H\times W} \tag{14}$$

where S is the output of Sobel Conv, $Att_L(\cdot)$ is the LA operation, $\odot$ denotes point-wise multiplication, and Sobel denotes sobel convolution. The combination of the GA and LA broadens the feature range. Moreover, the GA highlights important global features, and the LA enhances local detail features. They complement each other and improve the overall feature representation learning performance of the model.

In particular, the output of Sobel Conv interacts with the attention mechanism in two aspects: (1) In global attention, its transposed version generates channel-wise attention weights by using softmax to highlight edge-dependent cross-channel relationships; (2) In local attention, the raw output produces spatial attention weights through sigmoid to amplify features at edge locations.

3.5. Loss function

In this paper, the training of SSGAN consists of semi-supervised learning and generative adversarial learning. During the training stage of the encoder and decoder, results of labeled disease data are supervised by L_{ce}, and feature matching loss L_{fm} is used to minimize the difference between the pseudo label and the ground truth G. Lastly, L_{st} is employed to balance performance between the generator and discriminator. Thus, the loss function for SSGAN is formulated as follows:

$$L_{SSGAN}=L_{ce}+\beta_1 L_{fm}+\beta_2 L_{st} \tag{15}$$

where β_1 and β_2 are the balance parameters in the training process. The training process of SSGAN is displayed in Algorithm 1.

Algorithm 1. The training of SSSGAN.

Input: X_L: Labeled plant leaf diseases data
X_U: Unlabeled plant leaf diseases data
Output: Trained SSGAN
Define:
L_{ce}: Loss for labeled diseases data.
L_{fm}: Loss for ground truth and pseudo labels.
L_{st}: Loss for the balance between the generator and discriminator
L_{SSGAN}: Total loss for SSGAN
If training epoch < N:
Forward-propagation
Step 1. Generate the prediction and pseudo label of labeled and unlabeled data:
$Y_L = SS(X_L); Y_U = SS(X_U)$
Step 2. Generate feature map for generative adversarial learning:
$Y_t = G \oplus X_L; Y_f = SS(X_U) \oplus X_U$
Back-propagation
Step 1: Obtain loss for labeled diseases data:
$L_{ce} = \sum (Y_L - G)^2$
Step 2: Obtain feature matching loss:
$L_{fm} = \| E_{(X_L,G)\sim D^t}[D_k(G \oplus X_L)] - E_{(X_U,Y_U)\sim D^{st}}[D_k(SS(X_U) \oplus X_U)] \|$
Step 3: Obtain loss for self-training
$L_{st} = \begin{cases} -\sum_{h,w,c} Y_U \log SS(X_U), if D(SS(X_U)) \geqslant \lambda \\ 0, otherwise \end{cases}$
End

4. Experiments

4.1. Dataset description

To demonstrate the superiority of SSGAN in inspecting plant leaf diseases, the apple tree leaf diseases (Fang et al., 2024) and tomato leaf diseases datasets are considered (Hughes and Salathé, 2015) to perform experiments. The description of the two datasets is as follows.

1) Apple tree leaf disease dataset: The dataset was collected by Southwest University in China, which encompasses 1675 disease images of four distinct apple leaf diseases: Alternaria (417), Brown (411), Grey

(339), and Rust (508). Each diseased image was annotated with pixel labels.

2) Tomato leaf disease dataset: The dataset is sourced from Plant Village (Hughes and Salathé, 2015) and consists of 2451 tomato leaf diseases, which can be classified into late blight (612), spot (613), mold (641), and early blight (585) based on their characteristic. All the labels of disease images are at the pixel level.

The two datasets are both collected from real-world agriculture condition with various disease classes and high-quality labeling. The apple tree leaf disease dataset was selected for its comprehensive coverage of four distinct diseases with diverse background textures and scales, which poses significant challenges for automated segmentation. The tomato leaf disease dataset was chosen for its complex disease shapes and sizes, which poses great challenges for leaf diseases detection methods.

The images of the two datasets were uniformly preprocessed to a resolution of 160 × 256 pixels. The two datasets are partitioned into two segments: 80 % allocated for training and the remaining 20 % reserved for testing. Table 1 provides detailed information about the two datasets. It should be noted that labeled and unlabeled data account for 90 % and 10 % in the training set, respectively. Fig. 4 displays a selection of samples from them. It can be observed that the shape of diseased areas is mostly irregular. Proportion of diseased areas to leaf area is provided in Fig. 5. It is clear that most diseased areas occupy 0–20 %, while diseased areas over 80 % are only 2 % and 1 % in the two datasets. Moreover, the height and width ratio of the circumscribed rectangle of the diseased area are shown in Fig. 6. It can be seen that the maximum height and width ratio of the two datasets are 2.78 and 2.06, respectively. Additionally, the variances of the two datasets are 0.33 and 0.19, respectively. The statistical results above demonstrate that the diseased areas have complex geometric features, which pose a great challenge for diseased area detection.

4.2. Implementation details

The implementation of SSGAN is conducted by using PyTorch 1.10.0 and Python 3.8.13. Experiments are executed on a personal computer (PC) equipped with an Intel 4210R CPU and an NVIDIA RTX 3090 GPU, which boasts 24 GB of memory. SSGAN is trained employing the stochastic gradient descent (SGD) optimizer, utilizing 16 for batch-size, comprising 8 labeled and 8 unlabeled images. The training parameters are configured with an epoch of 250, weight decay of 0.95, and momentum of 0.001. During the training process, the random seed is fixed to ensure that the allocation of labeled and unlabeled data is the same when the experiments are conducted, which eliminates the influence of randomness in data partitioning on the experimental results. Moreover, the proportion of different classes of diseases in labeled and unlabeled selected based on the proportion of the original dataset, which is shown in Table 1.

4.3. Evaluation metrics

To evaluate the performance of SSGAN, four metrics commonly used in segmentation tasks are selected to perform comparative experiments with state-of-the-art methods, which are mean Intersection over Union (*mIoU*), Recall (*Rec*), Precision (*Pre*), and F1-score (*F1*). The formulation of these metrics is as follows:

$$mIoU = \frac{TP}{FP + FN + TP} \tag{16}$$

$$Pre = \frac{TP}{TP + FP} \tag{17}$$

$$Rec = \frac{TP}{TP + FN} \tag{18}$$

$$F1 = \frac{2 \times Pre \times Rec}{Pre + Rec} \tag{19}$$

where TP denotes pixels that have disease and are classified into disease, FP denotes pixels that are healthy but predicted as diseased, TN represents pixels with no disease and are predicted as healthy. FN represents pixels that are infected but predicted as healthy. To be specific, in the task of plant disease segmentation, mIoU represents the average ratio of the overlap between the segmented disease and the disease area in the ground truth. Precision represents the proportion of true disease pixels among the samples that are predicted as disease. Recall is the proportion of pixels that are actually with disease and correctly predicted as disease. F1-measure serves as a composite score that balances Precision and Recall.

To further analyze the effect of boundary feature, boundary precision (Pre^B), boundary recall (Rec^B), and boundary F1 ($F1^B$) are utilized. Specifically, Pre^B and Rec^B represent the degree of over-segmentation and under-segmentation, and $F1^B$ represents the degree of boundary localization in the results, which are as follows (Csurka et al., 2013):

$$Pr\,e^B = \frac{1}{|B_{ps}|} \sum_{z \in B_{ps}} \left[\left[d\left(z, B_{gt}\right) < \theta \right] \right] \tag{20}$$

$$Rec^B = \frac{1}{|B_{gt}|} \sum_{z \in B_{gt}} \left[\left[d\left(z, B_{ps}\right) < \theta \right] \right] \tag{21}$$

$$F1^B = \frac{2 \cdot Pr\,e^B \cdot Rec^B}{Rec^B + Pr\,e^B} \tag{22}$$

where B_{ps} and B_{gt} denote the boundary map of the predicted segmentation and the ground truth, respectively. $[[\cdot]]$ represents the Iverson bracket notation, $d(\cdot)$ denotes the Euclidean distance, θ represents the distance error tolerance, and z represents the pixel in the boundary of the disease.

4.4. Comparisons with state-of-the-art models

1) Statistical and visualization results analysis

Table 1
Details of the two datasets.

Dataset	Classes	Number (percentage)	Training	Testing	Labeled	Unlabeled
Tomato leaf	Late blight	612(25 %)	490	122	441	49
	Mold	641(26 %)	513	128	462	51
	Spot	613(25 %)	490	123	441	49
	Early blight	585(24 %)	468	117	421	47
Apple leaf	Alternaria	417(25 %)	334	83	301	33
	Brown	411(25 %)	329	82	296	33
	Grey	339(20 %)	271	68	244	27
	Rust	508(30 %)	406	102	365	41

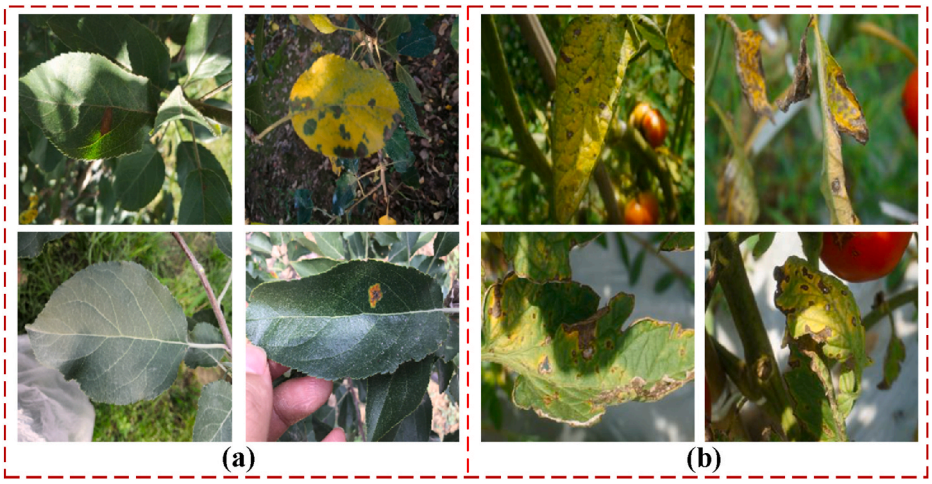


Fig. 4. Disease examples (a) Apple tree leaf diseases; (b) Tomato leaf diseases.

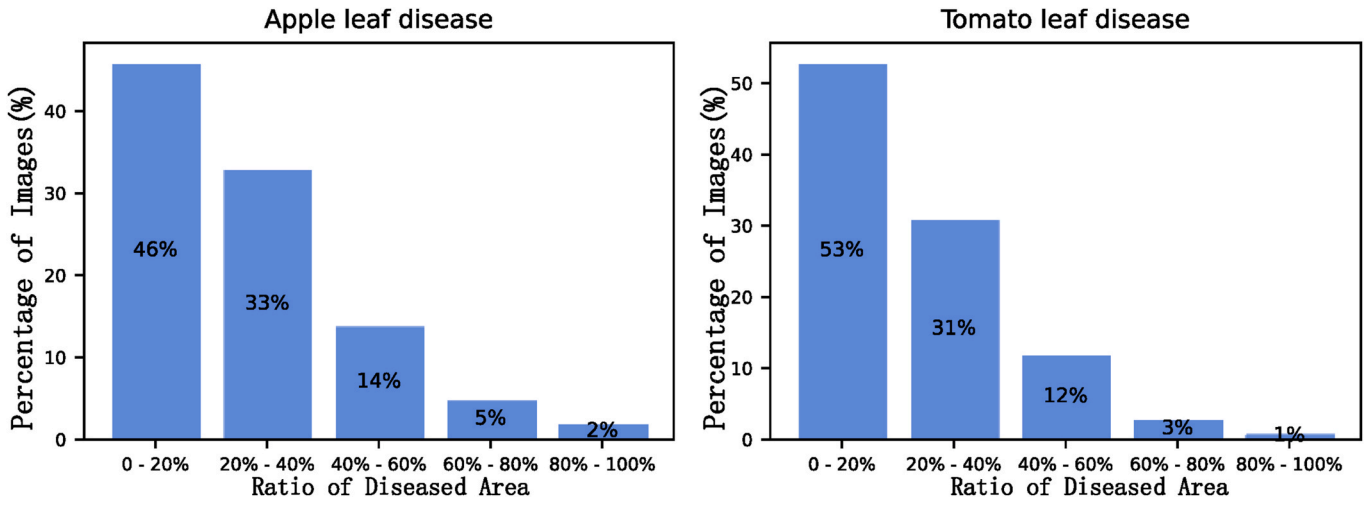


Fig. 5. Proportion of diseased areas to leaf area.

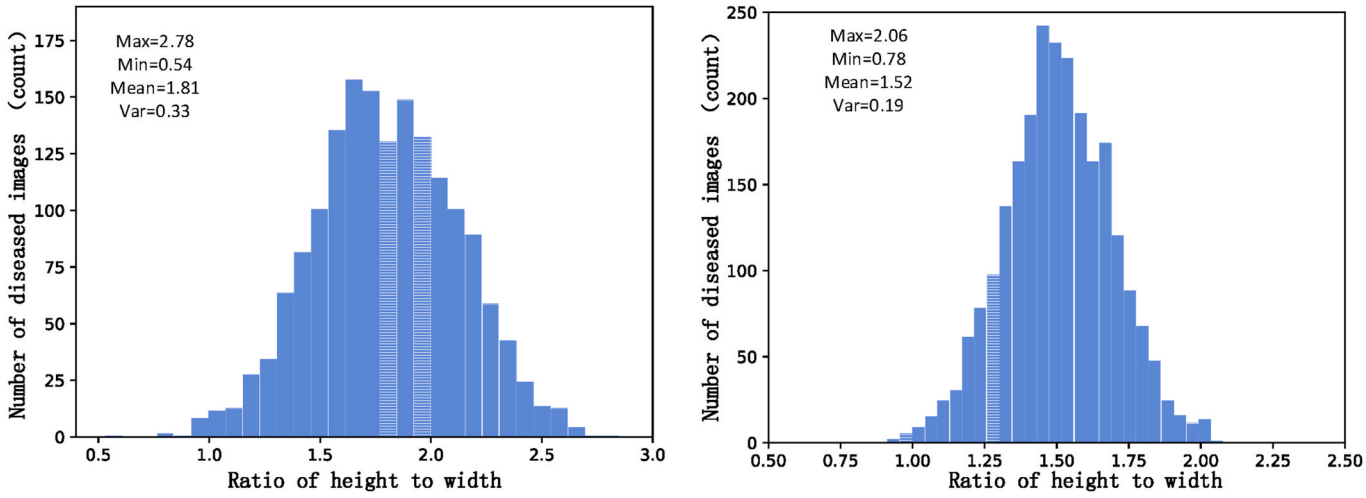


Fig. 6. Ratio of height/width of diseased area.

To illustrate the performance of SSGAN, five fully supervised methods (i.e., UNet (Ronneberger et al., 2015), Deeplabv3+ (Chen et al., 2017), UNet++ (Zhou et al., 2020), STV2 (Liu et al., 2022), Segformer (Xie et al., 2021), and MANet (Yut et al., 2024)), and five semi-supervised methods (i.e., CCT (Ouali et al., 2020), CPS (Chen et al., 2021a), CTCT (Luo et al., 2022b), CDMA (Zhong et al., 2023),

Table 2
Testing results of SSGAN with state-of-the-art methods.

Models	Data used		Apple leaf				Tomato leaf			
	Labeled	Unlabeled	Pre	Rec	F1	mIoU	Pre	Rec	F1	mIoU
UNet	10 %	0	75.70	71.33	73.45	62.32	84.71	81.73	83.16	73.62
Deeplabv3+			71.68	77.36	74.41	62.78	83.94	86.32	85.15	75.85
UNet++			71.03	77.82	74.27	62.61	86.23	84.76	85.42	74.73
STV2			71.65	78.71	75.01	63.32	84.52	88.10	86.25	77.12
Segformer			73.82	77.64	75.68	64.18	85.31	89.44	87.31	77.34
MANet			73.94	78.37	76.09	64.53	85.64	89.57	87.55	77.73
CCT	10 %	90 %	76.31	75.08	75.72	65.33	85.73	88.85	87.21	79.01
CPS			75.92	77.43	76.91	64.99	86.01	89.24	87.63	79.33
CTCT			78.93	75.39	77.24	66.73	86.25	88.72	87.57	79.61
CDMA			73.89	77.72	75.81	64.31	87.18	90.33	88.64	79.42
CorrMatch			74.73	76.92	75.83	65.72	87.57	89.49	88.53	79.54
DDFP			74.98	77.12	76.04	66.09	87.18	89.63	88.39	79.62
RMT			78.43	75.01	76.68	66.27	87.42	89.51	88.41	79.73
SSGAN			79.06	82.12	80.82	68.80	91.61	90.54	90.76	82.01

CorrMatch (Sun et al., 2024), DDFP (Wang et al., 2024), and RMT (Wang et al., 2022)) are considered to compare with SSGAN. It should be noted that fully-supervised models are trained only by 10 % of labeled images. Table 2 shows the testing results. It is clear that semi-supervised models have better performance than fully supervised methods, which demonstrates the superiority of semi-supervised learning methods. For the comparison with fully supervised methods, the best model, MANet, has mIoU of 64.53 % and 77.69 % on the two leaf disease datasets, respectively, which are 0.35 % and 0.39 % higher than the suboptimal fully supervised method Segformer, respectively. In particular, among the fully supervised models, UNet has the best precision of 75.70 % on the apple leaf dataset, and STV2 has the best recall of 78.71 % on the apple leaf dataset. While on the tomato leaf dataset, UNet++ has the best precision of 86.23 %, and MANet achieves the best recall of 89.57 %. For the comparison with semi-supervised methods, it is clear that SSGAN outperforms all other methods in the four metrics. It can be observed from Table 2 that on the apple leaf dataset, SSGAN achieves mIoU of 68.80 %, which is 2.07 % higher compared with the suboptimal method CTCT. While on the tomato leaf dataset, SSGAN achieves mIoU of 82.01 %, which is 2.28 % higher than the second-best method RMT.

Figs. 7 and 8 present some plant leaf disease samples and detection results on apple leaf and tomato leaf, respectively. For the apple leaf dataset, it is clear in Fig. 7 that SSGAN obtains the most accurate segmentation results. In particular, in the fourth row where the shape of disease regions is simply regular, most methods can segment the diseased area. However, when diseases have fuzzy edges shown in the first and second rows, these methods, e.g., UNet, Deeplabv3, UNet++, STV2, Segformer, MANet cannot detect the diseases accurately. Due to the BFAM, SSGAN can recognize the diseased regions that have unclear boundaries. In the third row, where the background noise is severe, these methods, like Segformer, CPS, CTCT, CorrMatch, DDFP incompletely or wrongly recognize the diseases. However, SSGAN can still provide good results, which can be attributed to the effectiveness of BNSM. For the tomato leaf dataset, it can be observed from the second and third row that UNet and Segformer fail to segment the diseased area due to their complex shapes, while SSGAN achieves the best segmentation results. That is because BFAM enables to network to extract more edge features. Moreover, the samples in the tomato leaf dataset all have strong background noise, which makes STV2 and CCT to fail to segment all the diseased areas. SSGAN can still have better performance because BNSM can effectively reduce the adverse influence of the background noise. Thus, it can be concluded that SSGAN is better than other methods.

To further demonstrate the superiority of SSGAN, T-test (West, 2021) experiments on the two datasets are conducted. The P value in the t-test of SSGAN compared with that of other methods is shown in Tables 3 and 4. It can be seen that all the P values are smaller than 0.05, which indicates that SSGAN is significantly superior to other methods.

2) Analysis of complexity

To further verify the computational complexity of SSGAN, Floating Point Operations Per Second (FLOPs) and Frames Per Second (FPS) are used to compare SSGAN with other methods. The results are shown in Table 5. It can be seen that Segformer has the lowest FLOPs, and the FLOPs of SSGAN are lower than those of most other methods. It is clear that SSGAN has an FPS of 41, which can satisfy the need for real-time segmentation. Although UNet has an FPS of 43, SSGAN achieves better accuracy. It can be concluded that SSGAN can achieve high segmentation performance while maintaining comparatively high inference speed.

3) Visualization of feature extraction and feature selection

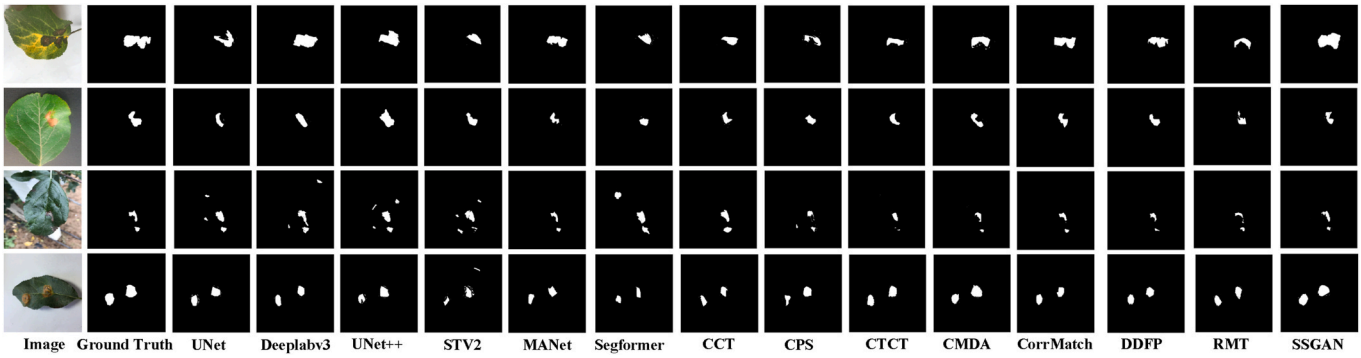


Fig. 7. Visualization comparison on the apple tree leaf dataset.

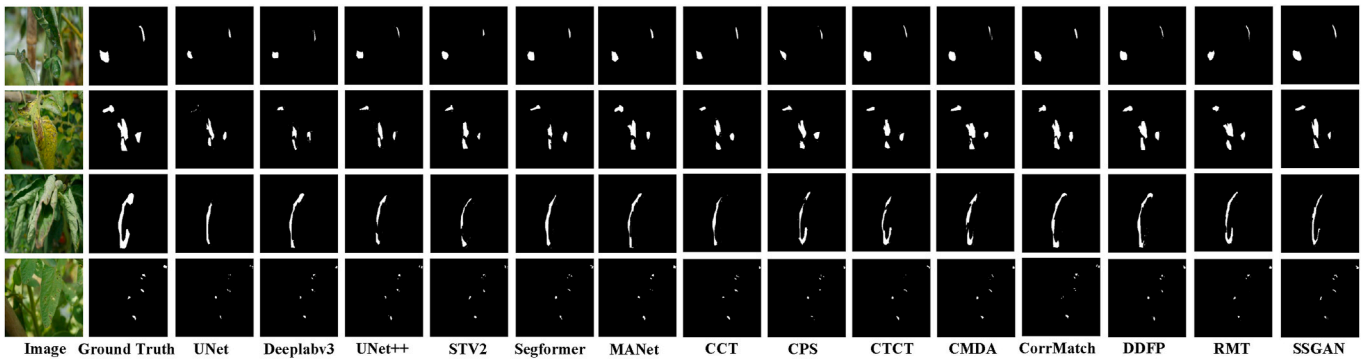


Fig. 8. Visualization comparison on the tomato leaf dataset.

Table 3
T-test on apple leaf datasets.

Methods	UNet	Deeplabv3+	UNet++	STV2	Segformer	MANet	CCT	CPS	CTCT	CDMA	DDFP	CorrMatch	RMT
Pre	1.29e-17	2.47e-23	5.63e-23	4.55e-24	7.57e-21	3.81e-19	8.64e-15	7.95e-16	1.78e-01	2.92e-20	4.53e-17	6.66e-17	1.65e-05
Rec	5.59e-27	3.26e-21	1.58e-21	5.54e-17	1.51e-20	7.58e-17	3.56e-24	1.67e-21	3.25e-23	2.34e-20	4.98e-19	3.63e-19	2.19e-24
F1	8.05e-25	1.92e-23	2.77e-24	1.14e-23	3.18e-22	3.18e-18	1.70e-21	2.48e-20	2.83e-19	1.92e-21	1.15e-18	2.75e-18	3.23e-21

To visualize the process of feature extraction and feature selection of SSGAN, t-SNE (Cimpoesu et al., 2025) and heatmaps are generated. For the feature extraction, it can be seen from Fig. 9 that the features of different diseases are generally confused after ResNet34. After BFAM, the features are separated for recognition of each class of disease. Moreover, the features of diseases have higher discriminability after BNSM. It indicates that BFAM and BNSM have a positive effect on the feature extraction process. As to the feature selection, Grad-CAM (Selvaraju et al., 2017) is used to generate heatmaps, and the red area means the features that have higher weight. It can be seen from Fig. 10 that the feature intensity is not evident in the diseased area after ResNet34. After BFAM, the red areas cover more diseased parts. Additionally, red areas are more evident after BNSM. Thus, it can be concluded that BFAM and BNSM enable the network to better implement feature selection.

4.5. Ablation analysis

1) Ablation study of GAN, BFAM, and BNSM

In order to verify the effectiveness of each component in SSGAN, ablative studies are performed on the two datasets. Table 6 and Fig. 11 show the result of four metrics and their percentage drops under different settings, where M1, M2, and M3 denote the model when GAN, BFAM, and BNSM are removed, respectively.

It can be observed that the performance of SSGAN decreases when GAN, BFAM, and BNSM are removed. Only the combination of all modules deployed in our network enables the best performance, which achieves 68.80 % and 82.01 % mIoU in the two datasets, respectively. In particular, mIoU decreases by 1.55 % and 1.62 % when the GAN component is removed, which indicates that GAN enables the network to learn disease features under limited data. Meanwhile, due to the boundary feature enhancement by BFAM, mIoU increases by 0.93 % and 1.38 %. When the BNSM is removed, mIoU decreases by 0.68 % and 1.14 %. That is because the noise from the background significantly affects the performance of SSGAN without the presence of BNSM. To better demonstrate the effectiveness of BNSM, heatmaps are generated in Fig. 12. It can be observed that the background noise is severe without BNSM. The noise is alleviated when BNSM is added, which shows the effectiveness of BNSM in reducing noise and enhancing disease-specific features. According to the results from the ablation experiments, it can be concluded that it makes a substantial contribution to improving disease segmentation.

To evaluate whether BNSM influences the boundary localization, ablation studies are conducted based on the metric Pre^B, $Recall^B$, and $F1^B$. The results are shown in Table 7. It is clear that Pre^B, $Recall^B$, and

Table 4
T-test on tomato leaf datasets.

Methods	UNet	Deeplabv3+	UNet++	STV2	Segformer	MANet	CCT	CPS	CTCT	CDMA	DDFP	CorrMatch	RMT
Pre	1.87e-23	1.52e-23	3.75e-20	2.04e-22	6.20e-22	6.44e-20	1.43e-21	3.53e-23	3.43e-22	1.39e-19	9.59e-17	6.85e-18	1.68e-19
Rec	8.22e-25	9.06e-19	2.22e-21	9.47e-15	1.12e-09	1.14e-04	1.71e-12	1.24e-10	3.95e-13	0.01	3.07e-05	2.98e-04	4.91e-08
F1	1.99e-25	3.70e-23	5.29e-22	8.58e-22	4.88e-20	7.68e-16	7.88e-20	1.29e-18	1.48e-19	3.92e-16	4.02e-13	1.55e-13	3.56e-17

Table 5
Testing results of computational complexity.

Method	UNet	Deeplabv3+	UNet++	STV2	Segformer	MANet	CCT	CPS	CTCT	CDMA	DDFP	CorrMatch	RMT	SSGAN
FLOPs(G)	70.42	60.45	127.08	79.15	**2.49**	82.47	59.34	72.47	84.13	54.32	64.32	72.47	76.64	64.53
FPS	**43**	39	40	41	42	42	40	37	39	40	39	37	42	41

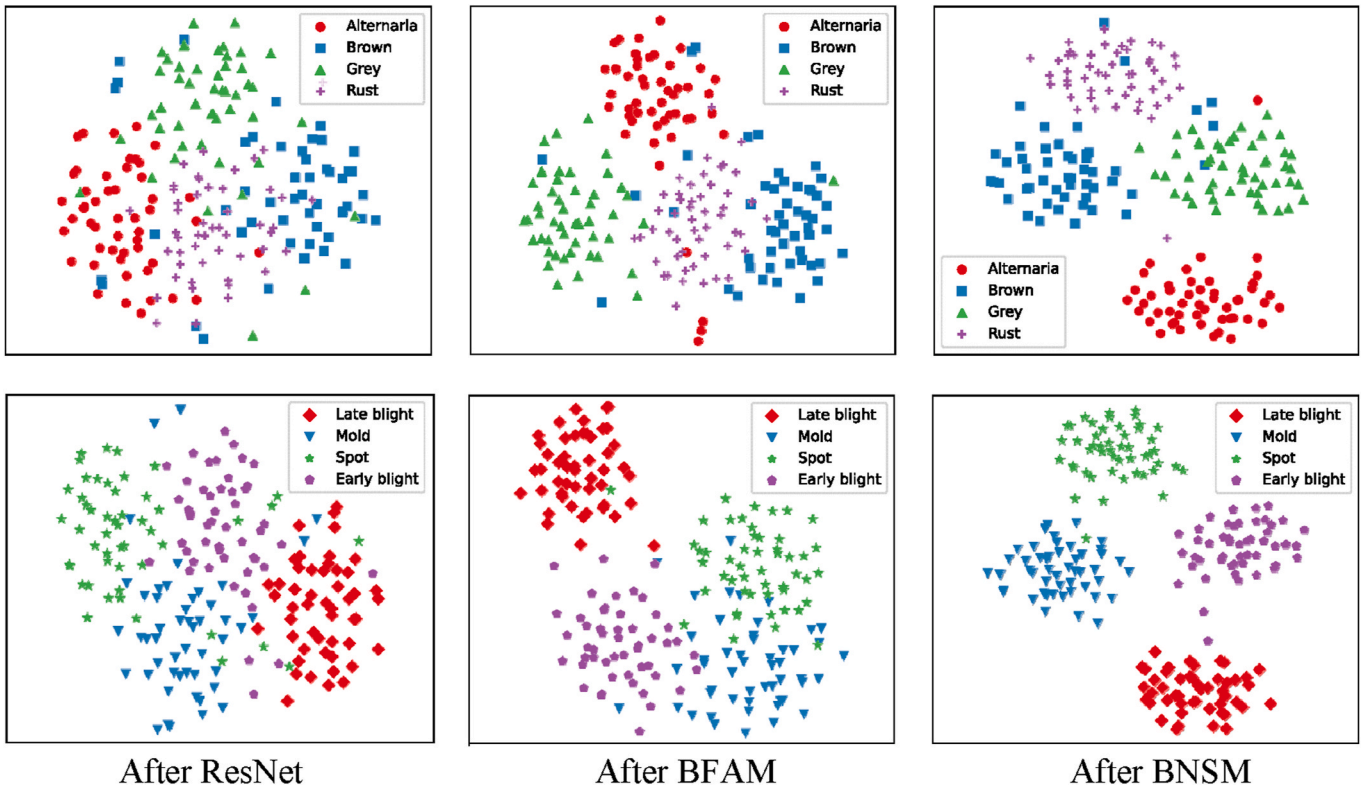


Fig. 9. Visualization of feature extraction process.

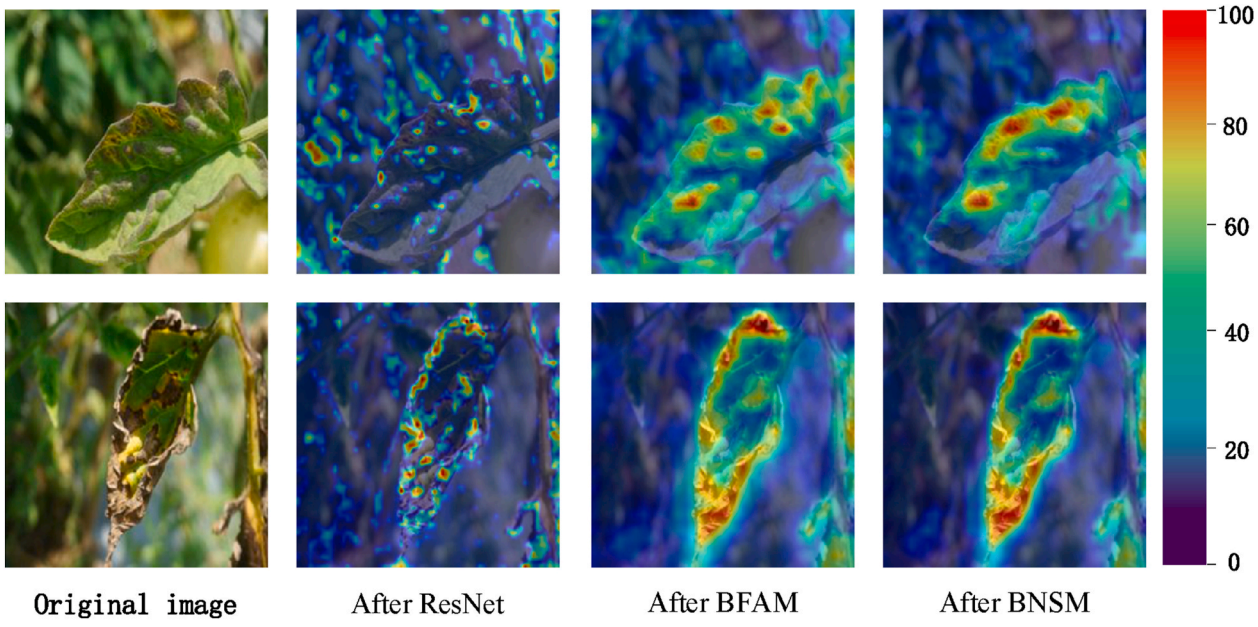


Fig. 10. Visualization of feature selection process.

Table 6
Results of ablative study (%).

Methods	GAN	BFAM	BNSM	Apple tree leaf				Tomato tree leaf			
				Precision	Recall	F1	mIoU	Precision	Recall	F1	mIoU
SSGAN	✓	✓	✓	79.06	82.12	80.82	68.80	91.61	90.54	90.76	82.01
M1		✓	✓	75.18(-3.88)	81.03(-1.09)	78.29(-2.53)	67.35(-1.45)	89.43(-2.18)	86.24(-4.30)	87.81(-2.95)	80.39(-1.62)
M2	✓		✓	75.44(-3.62)	81.52(-0.60)	78.62(-2.20)	67.87(-0.93)	89.82(-1.79)	86.82(-3.72)	88.03(-2.73)	80.63(-1.38)
M3	✓	✓		75.72(-3.34)	81.87(-0.25)	78.94(-1.88)	68.12(-0.68)	90.32(-1.29)	87.13(-3.41)	88.70(-2.06)	80.87(-1.14)

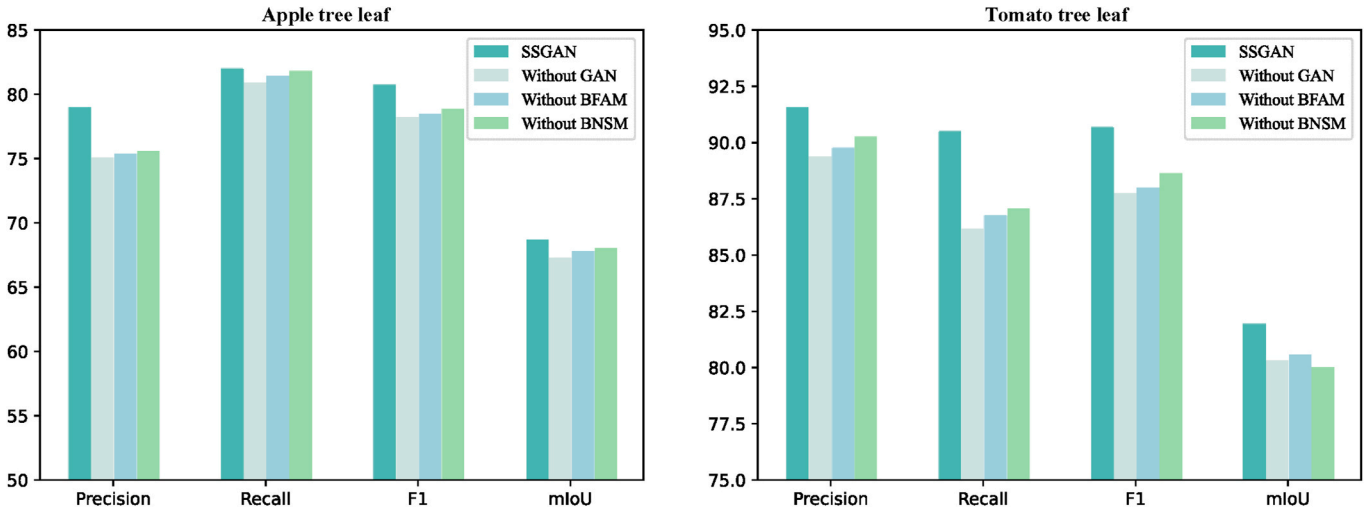


Fig. 11. Comparative results of ablation analysis.

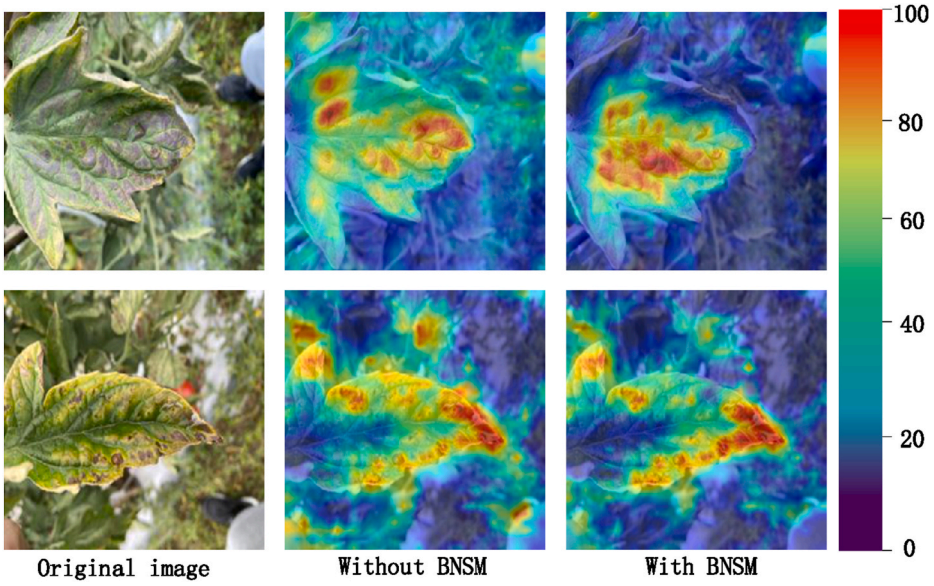


Fig. 12. Heatmap visualization of BNSM.

$F1^B$ decrease sharply when BNSM and BFAM are removed. In particular, the lower Pre^B and $Recall^B$ represents more over-segmentation and under-segmentation. It can be seen from Table 7 that SSGAN achieves the best Pre^B and $Recall^B$. Thus, it can be concluded that SSGAN can segment the diseased area accurately. Moreover, as to $F1^B$ which denotes the accuracy of boundary localization, higher $F1^B$ means better boundary localization. It can be observed that when BFAM is removed, $F1^B$ decreases by 2.14 % and 4.97 % on the two datasets, respectively. Additionally, there is a decline in $F1^B$ when BNSM is removed. Additionally, the overlap visualizations of segmentation results are displayed in Fig. 13. It can be observed that SSGAN has the highest overlapping ratio with ground truth while the performance of edge detection decreases when BFAM is removed. Thus, it can be concluded that BNSM can also contribute to the boundary localization to some extent.

Table 7
Results of boundary metrics (%).

Method	Apple leaf disease			Tomato leaf disease		
	Pre^B	Rec^B	$F1^B$	Pre^B	$Recall^B$	$F1^B$
SSGAN	64.28	69.82	67.08	74.52	72.54	73.68
Without BFAM	61.36	68.72	64.94	70.38	68.93	68.71
Without BNSM	63.57	69.16	66.36	73.96	71.72	72.88

2) Ablation study of different decoding modules

To evaluate the performance of SSGAN under different decoding modules, PSP (Deng et al., 2021), SPP (Lin et al., 2024), EMCA (Rahman et al., 2024), and FPA (Fang et al., 2023a) are selected to conduct experiments. Table 8 shows the results. It can be seen from Table 8 that ASPP has the best performance on the two datasets on all four metrics. In particular, when using ASPP, the mIoU improves by 0.39 % and 0.18 % higher than the suboptimal module EMCA on the two datasets, respectively. Thus, ASPP is selected as the decoding module in this paper.

4.6. Sensitive analysis

1) Batch size and label percent

The sensitive experiments are conducted to determine the proportion

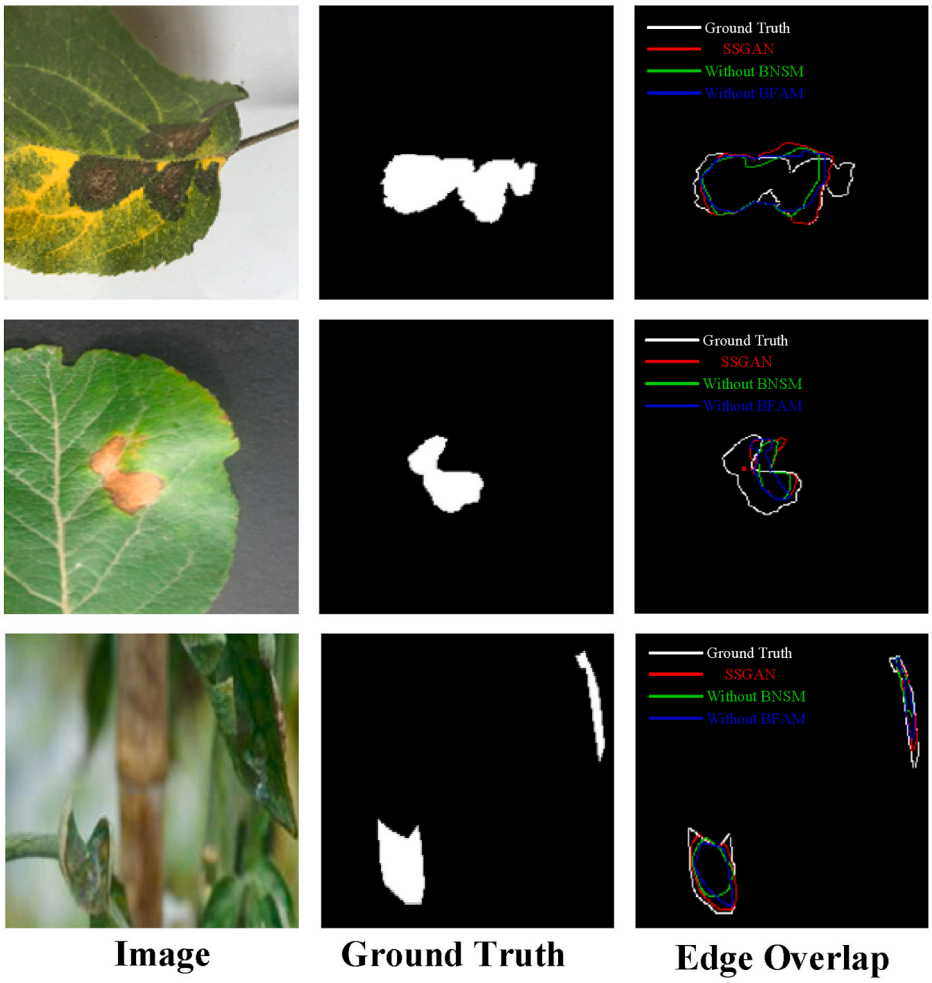


Fig. 13. Overlap visualization of segmentation results.

Table 8
Results of different decoding modules (%).

Methods	Decoding module	Apple leaf				Tomato leaf			
		Pre	Rec	F1	mIoU	Pre	Rec	F1	mIoU
SSGAN	ASPP	76.06	82.12	78.81	68.80	91.61	87.54	89.63	82.01
	PSP (Deng et al., 2021)	75.90	81.75	78.42	68.16	91.32	87.13	88.95	81.54
	SPP (Lin et al., 2024)	75.32	81.19	78.57	67.92	91.08	86.52	88.84	81.26
	EMCA (Rahman et al., 2024)	75.92	81.84	78.59	68.41	91.49	87.32	89.05	81.83
	FPA (Fang et al., 2023a)	75.62	81.52	78.29	68.12	91.19	86.94	88.76	81.42

of the training batch size (bs) and labeled data. Testing results are shown in Tables 9 and 10. The testing is conducted on the training set of the apple tree leaf dataset. Compared to fully-supervised approaches, semi-supervised methods consistently demonstrate superior performance, and SSGAN shows the best performance among all the semi-supervised methods. It can be seen from Table 7 that mIoU increases when the labeled percentage becomes higher. When the labeled percent increases to 10 %, all the methods have the most significant improvement in mIoU, with UNet of 3.01 %, Deeplabv3+ of 2.86 %, UNet++ of 2.75 %, STV2 of 3.17 %, Segformer of 3.34 %, CCT of 2.94 %, CPS of 3.62 %, CTCT of 2.98 %, CDMA of 2.60 %, RMT of 2.8 %, and SSGAN of 3.49 %. While from 10 % to 15 %, SSGAN increases the mIoU by 2.16 %. When

Table 9
Sensitive analysis of labeled percent (%).

Method	UNet	Deeplabv3+	UNet++	STV2	Segformer	MANet	CCT	CPS	CTCT	CDMA	CorrMatch	DDFP	RMT	SSGAN
5 %	59.31	59.92	59.86	60.15	60.84	61.14	62.39	61.14	63.54	61.58	62.06	62.48	63.47	65.31
10 %	62.32	62.78	62.61	63.32	64.18	64.53	65.33	64.99	66.73	64.31	65.72	66.09	66.27	68.80
15 %	63.47	64.13	63.90	64.53	65.47	65.62	66.64	66.24	67.96	65.62	66.95	67.39	67.57	70.96
20 %	64.02	64.62	64.47	64.91	66.02	66.13	67.29	66.78	68.54	66.17	67.49	67.85	68.03	71.43
25 %	64.39	64.96	64.81	65.17	66.59	66.87	67.72	67.23	69.08	66.52	67.82	68.05	68.36	71.88
30 %	64.53	65.20	65.03	65.32	66.83	66.92	67.99	67.48	69.27	66.71	67.90	68.14	68.52	71.97

Table 10
Sensitive analysis of batch size (%).

Number of labeled/unlabeled	2/2	4/4	8/8	16/16
mIoU	67.63	68.15	68.80	68.47
Pre	75.22	75.43	76.06	75.69
Rec	81.53	81.93	82.12	81.96
F1	78.43	78.72	78.81	78.69

the labeled proportion is increased to 20 %, the improvement of mIoU declines compared with 5 %–10 % and 10 %–15 %, where the highest improvement is 0.65 % by CCT. SSGAN achieves the best mIoU of 71.97 % when the labeled percent is 30 %, which is 3.45 % higher than the suboptimal method RMT. Additionally, it is worth noting that the act of labeling in practical applications is time-consuming. Therefore, 10 % is chosen as the label proportion to balance accuracy and detection time. As to the batch size of training, the range of the batch sizes used in the experiments is decided based on literature (Bai et al., 2023). It can be seen from Table 10 that SSGAN achieves the best performance with mIoU of 68.80 %, Precision of 76.06 %, Recall of 82.12 %, and F1 of 78.81 %. The mIoU decreases by 1.17 %, 0.65 %, and 0.33 % compared with 2/2, 4/4, and 16/16, respectively. Thus, 8 labeled and 8 unlabeled images are selected as the training batch size.

2) Threshold λ

To evaluate the influence of λ, different λ are used to conduct experiments. The results are shown in Table 11. The range of λ is based on literature (Chen et al., 2021b). It is clear that when the λ is 0.9, SSGAN achieves the best performance on the two datasets. In particular, mIoU decreases 0.38 % on the apple leaf dataset when choosing 0.85 as λ. While on the tomato leaf dataset, mIoU decreases 0.28 % with the suboptimal λ 0.80. Thus, 0.9 is selected as the parameter λ in the experiments.

4.7. Extension study

1) Network pruning

To illustrate the performance of SSGAN in resource-constrained environments, the four pruning methods, i.e., DepGraph (Fang et al., 2023b), DAC (Zhuang et al., 2018), FPGM (He et al., 2019), and PCES (Guo and Potkonjak, 2017), are used to conduct experiments. The mIoU is displayed in Table 12. It is clear that the mIoU decreases when the pruning techniques are implemented. DAC has the best pruned proportion of 10.8 % while its mIoU decreases by 1.07 % and 1.37 % on the two datasets, respectively. The suboptimal pruning approach FPGM achieves 9.4 % parameter reduction, and its mIoU declines 1.09 % and 0.80 % on the two datasets. It can be concluded from Table 12 that the four pruning techniques can effectively reduce the parameters while maintaining comparatively high performance.

2) Domain adaptation

To improve the performance of SSGAN across different datasets, four representative domain adaptation loss functions, Max mean discrepancy loss (MMDL) (Schwendemann et al., 2021), adversarial loss (AL) (Ju et al., 2025), wasserstein divergence loss (WDL) (Si et al., 2024), and reconstruction loss (RL) (Ji et al., 2025), are employed to conduct experiments. The results are obtained from cross-domain testing, which is shown in Table 13. Apple-Tomato denotes that the training is performed with the apple leaf dataset and 10 % tomato dataset, and the testing is performed on the tomato leaf dataset. Tomato-Apple denotes that the training is performed with the tomato leaf dataset and 10 % apple leaf dataset, and the testing is performed on the apple leaf dataset. The first line denotes that the training is performed on one dataset and then is directly tested on the other dataset. For the comparison with the setting in Table 2, mIoU decreases by 10.65 % and 10.29 % on the two datasets, respectively, which illustrates the adverse influence of domain shift. Thus, it can be concluded that the performance of SSGAN degrades when confronted with the unseen dataset. Moreover, it can be observed that the performance of SSGAN improves when domain adaptation techniques are implemented. In particular, mIoU increases by 3.76 % and 1.27 % on the two datasets, respectively, when AL is added. The second-best loss function, WDL, improves the mIoU by 3.38 % and 0.91 % on the two datasets, respectively. Thus, it can be concluded that domain adaptation technique enables SSGAN to achieve good generalization across different datasets.

Table 12
Testing results of different pruning methods (%).

Method	Pruned	Parameters	mIoU	
			Apple	Tomato
Original	0 %	2.16e+7	68.80	82.01
DepGraph	7.6 %	1.99e+7	68.06	81.57
DAC	10.8 %	1.93e+7	67.73	80.64
FPGM	9.4 %	1.95e+7	67.91	81.21
PCES	7.2 %	2.01e+7	68.25	81.65

3) Imbalanced dataset

To evaluate the performance of SSGAN under imbalanced data, the proportion of the two datasets is adjusted to make the two datasets more imbalanced, which is shown in Table 14. The testing results are shown in Table 15.

It can be seen that the performance of SSGAN degrades a lot when the two datasets are imbalanced while SSGAN still outperforms other methods (see Table 16). In particular, the mIoU of SSGAN on the two datasets decreases by 6.49 % and 4.64 % compared with balanced data, respectively. For the comparison with other methods, the mIoU of SSGAN is 0.84 % higher than the second-best method RMT on the apple leaf dataset. Moreover, on the tomato leaf dataset, the suboptimal method is CDMA, with 2.88 % lower than SSGAN. It can be concluded that imbalanced data can have an adverse effect on the performance of SSGAN.

4) Role of pseudo labels

In semi-supervised plant leaf disease segmentation, pseudo labels

Table 11
Sensitive analysis of λ (%).

λ	Apple leaf				Tomato leaf			
	Pre	Rec	F1	mIoU	Pre	Rec	F1	mIoU
0.75	74.33	80.91	77.47	67.29	90.54	86.25	88.34	81.29
0.80	75.49	81.46	78.36	68.14	91.29	86.64	87.78	81.73
0.85	75.93	81.85	78.78	68.42	91.03	87.11	89.14	81.65
0.90	76.06	82.12	78.81	68.80	91.61	87.54	89.63	82.01
0.95	75.64	81.49	78.46	68.10	90.84	86.48	88.60	81.25

Table 13
Testing results of domain adaptation (%).

Method	Apple-Tomato				Tomato-Apple			
	Pre	Rec	F1	mIoU	Pre	Rec	F1	mIoU
SSGAN	64.12 ± 0.04	72.75 ± 0.02	68.09 ± 0.12	58.16 ± 0.03	79.32 ± 0.01	82.57 ± 0.14	80.83 ± 0.33	71.72 ± 0.28
SSGAN + MMD	65.25 ± 0.09	73.08 ± 0.01	69.11 ± 0.02	60.32 ± 0.04	79.97 ± 0.04	82.87 ± 0.03	81.04 ± 0.01	72.18 ± 0.01
SSGAN + AL	66.24 ± 0.03	73.82 ± 0.01	69.95 ± 0.03	61.92 ± 0.01	81.02 ± 0.02	83.97 ± 0.16	82.54 ± 0.02	72.99 ± 0.04
SSGAN + WDL	66.03 ± 0.52	73.51 ± 0.03	69.83 ± 0.01	61.54 ± 0.02	80.63 ± 0.01	83.42 ± 0.03	82.12 ± 0.13	72.63 ± 0.02
SSGAN + RL	65.57 ± 0.18	73.38 ± 0.01	69.42 ± 0.10	60.84 ± 0.03	80.32 ± 0.01	83.08 ± 0.26	81.62 ± 0.01	72.37 ± 0.01

Table 14
Data quantity of the imbalanced dataset.

Tomato leaf			Apple leaf		
Type	Quantity	Proportion	Type	Quantity	Proportion
Late blight	500	36 %	Alternaria	400	29 %
Mold	400	29 %	Brown	300	21 %
Spot	300	21 %	Grey	200	14 %
Early blight	200	14 %	Rust	500	36 %

effectively enhance overall performance by utilizing unlabeled data to expand the data scale and enrich features for good generalization. Moreover, it exerts a regularization effect to prevent overfitting, which leverages a self-training mechanism to improve image segmentation accuracy. To evaluate the influence of pseudo labels, comparative experiments are conducted. The testing results are shown in Table 16. Firstly, a benchmark model named B1 is trained using only 10 % labeled data. Secondly, pseudo labels from 90 % of the unlabeled data are generated by using the model. Then the 90 % unlabeled data and their corresponding pseudo labels are regarded as data with labels. Then, 10 % labeled data and 90 % unlabeled data are mixed to train the other model B2. It should be noted that the two tests have the same hyperparameter settings such as learning rate, optimizer, number of training epochs, and testing set. It can be observed that the mIoU of B2 is 3.37 % and 3.56 % higher than B1, which demonstrates the superiority of B2. Thus, it can be concluded that pseudo labels have a positive effect on the performance of the proposed model.

5) Explainable visualization by DeepSHAP

To improve the explainability of SSGAN, DeepSHAP (Saranya and Narayan, 2024) is utilized to visualize the feature and decision process of the model. The results are shown in Fig. 14. Particularly, the red areas represent the regions with high weight. It can be seen from Fig. 14 that the areas covered with red are mostly diseased areas, which verifies the feature extraction performance of SSGAN.

6) Bayesian optimization of SSGAN

To explore the optimal hyperparameter of SSGAN, Bayesian optimization (Song and Jiang, 2025) is employed to conduct experiments. Specifically, three critical hyperparameters, drop-out ratio (dr), learning rate (lr), and batch size (bs), are selected to obtain their optimal value. The results are presented in Tables 17 and 18, respectively. It can be seen that after Bayesian optimization, the bs are the same as the original value, while the dr and lr are changed into 0.25 and 0.05, 0.2 and 0.01 on the two datasets, respectively. Moreover, the mIoU on the two datasets improve by 0.23 % and 0.18 %, which demonstrates the effectiveness of Bayesian optimization.

5. Conclusion

In this paper, a semi-supervised segmentation model, i.e., SSGAN, is developed for plant leaf disease detection. Firstly, semi-supervised generative adversarial learning is developed to take advantage of labeled and unlabeled disease data to get more semantic features. Secondly, to enhance the feature discrimination of disease boundary, a boundary feature attention module is developed by utilizing global and local attention to obtain edge cues. Lastly, to alleviate interference from background, a background noise suppression module is developed to improve the differentiation between disease-affected areas and the surrounding leaf background. The effectiveness of SSGAN is verified on two leaf diseases datasets. The testing results show that SSGAN effectively detects plant leaf diseases. Ablation study has also shown that GAN, BFAM, and BNSM all contributed to the disease inspection performance of SSGAN. However, the two datasets used in this study are captured under high-quality lighting environments. The performance of SSGAN may degrade when confronted with extreme lighting conditions

Table 15
Testing results of the imbalanced dataset.

Methods	Apple leaf				Tomato leaf			
	Pre	Rec	F1	mIoU	Pre	Rec	F1	mIoU
CCT	72.47	72.19	72.31	59.89	80.15	83.71	81.75	73.25
CPS	72.92	71.86	72.53	59.04	81.21	84.62	82.66	73.92
CTCT	72.93	72.74	72.94	61.36	81.43	83.53	82.39	73.64
CDMA	70.94	72.92	72.03	58.13	82.40	85.12	83.93	74.49
RMT	72.35	73.01	72.68	61.47	82.63	84.41	83.33	74.06
SSGAN(Imbalanced)	74.32	76.75	75.77	62.31	84.36	85.23	84.91	77.37
SSGAN(Balanced)	79.06	82.12	80.82	68.80	91.61	90.54	90.76	82.01

Table 16
Testing results of the influence of pseudo labels.

Methods	Apple leaf				Tomato leaf			
	Pre	Rec	F1	mIoU	Pre	Rec	F1	mIoU
B1	74.14	77.91	75.98	64.40	85.79	88.75	86.15	77.89
B2	76.85	79.53	78.18	67.47	88.07	89.06	88.56	81.45

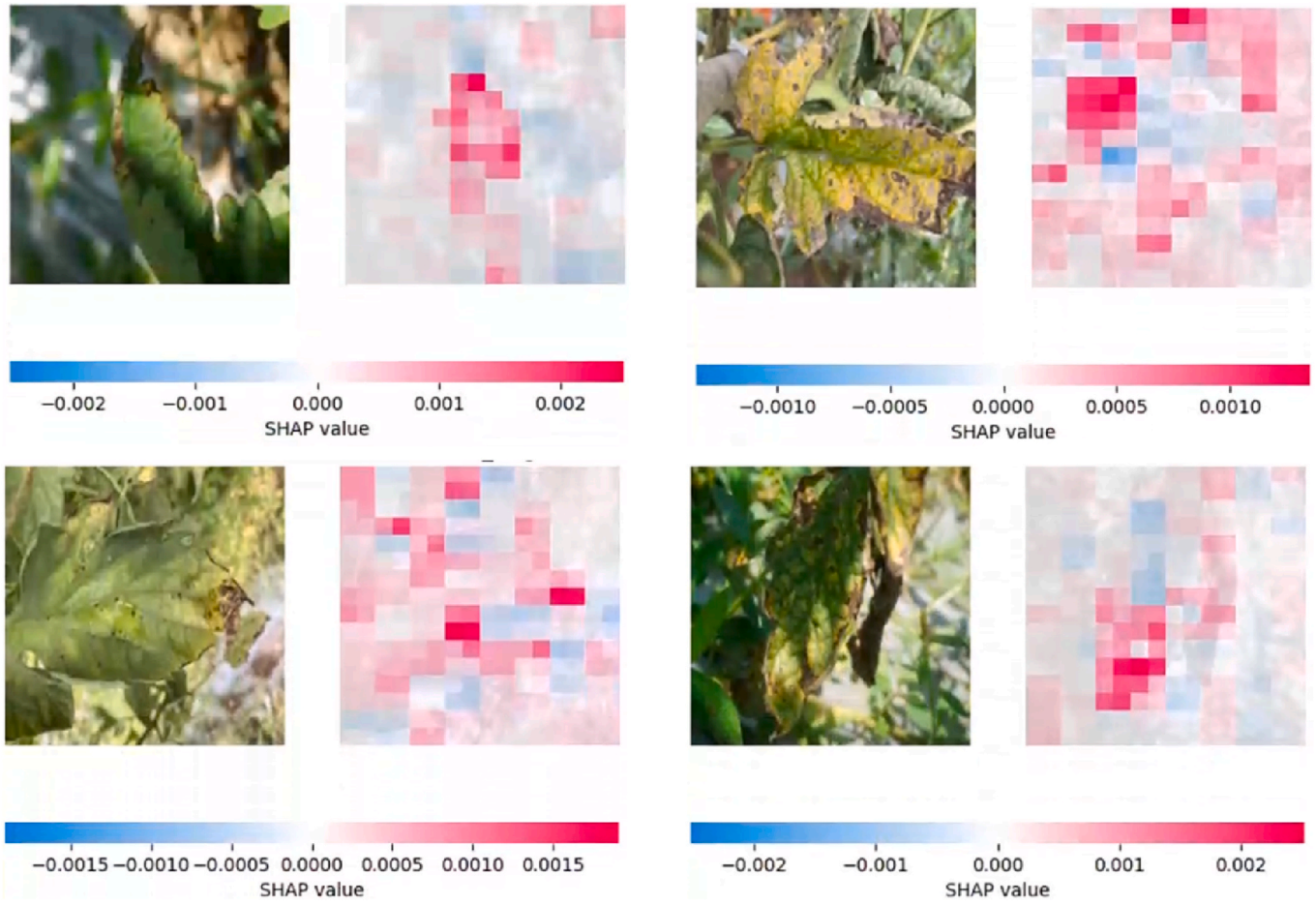


Fig. 14. Visualization of DeepSHAP.

Table 17
Bayesian optimization on apple leaf dataset.

Model	Hyperparameter			Pre	Rec	F1	mIoU
	dr	lr	bs				
SSGAN	0.3	0.1	16	79.06	82.12	80.82	68.80
Optimized	0.25	0.05	16	79.42	82.39	80.87	69.03

Table 18
Bayesian optimization on tomato leaf dataset.

Model	Hyperparameter			Pre	Rec	F1	mIoU
	dr	lr	bs				
SSGAN	0.3	0.1	16	91.61	90.54	90.76	82.01
Optimized	0.2	0.01	16	91.83	90.62	91.24	82.19

such as darkness and shadow. Moreover, SSGAN can achieve great performance under a large quantity of images with limited labels. In real-world deployment scenarios, diseased samples are scarce, which poses a challenge to plant leaf disease detection. Future research will concentrate on applying SSGAN under varying lighting, camera angles, and more severe background clutter. In addition, few-shot learning will be explored for limited samples to enhance plant leaf disease detection.

6. Source of the dataset

Apple tree leaf: https://www.scidb.cn/en/detail?dataSetId=0e1f57004db842f99668d82183afd578&dataSet.

Tomato leaf: https://aistudio.baidu.com/datasetdetail/245434.

7. Code of Grad-CAM

We conducted the visualization of Grad-CAM based on the code from https://github.com/frgfm/torch-cam.

CRediT authorship contribution statement

Lixiang Zhao: Writing – original draft, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Jun Hao:** Validation, Project administration, Funding acquisition. **Demin Li:** Supervision, Project administration, Investigation. **Jianbo Yu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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